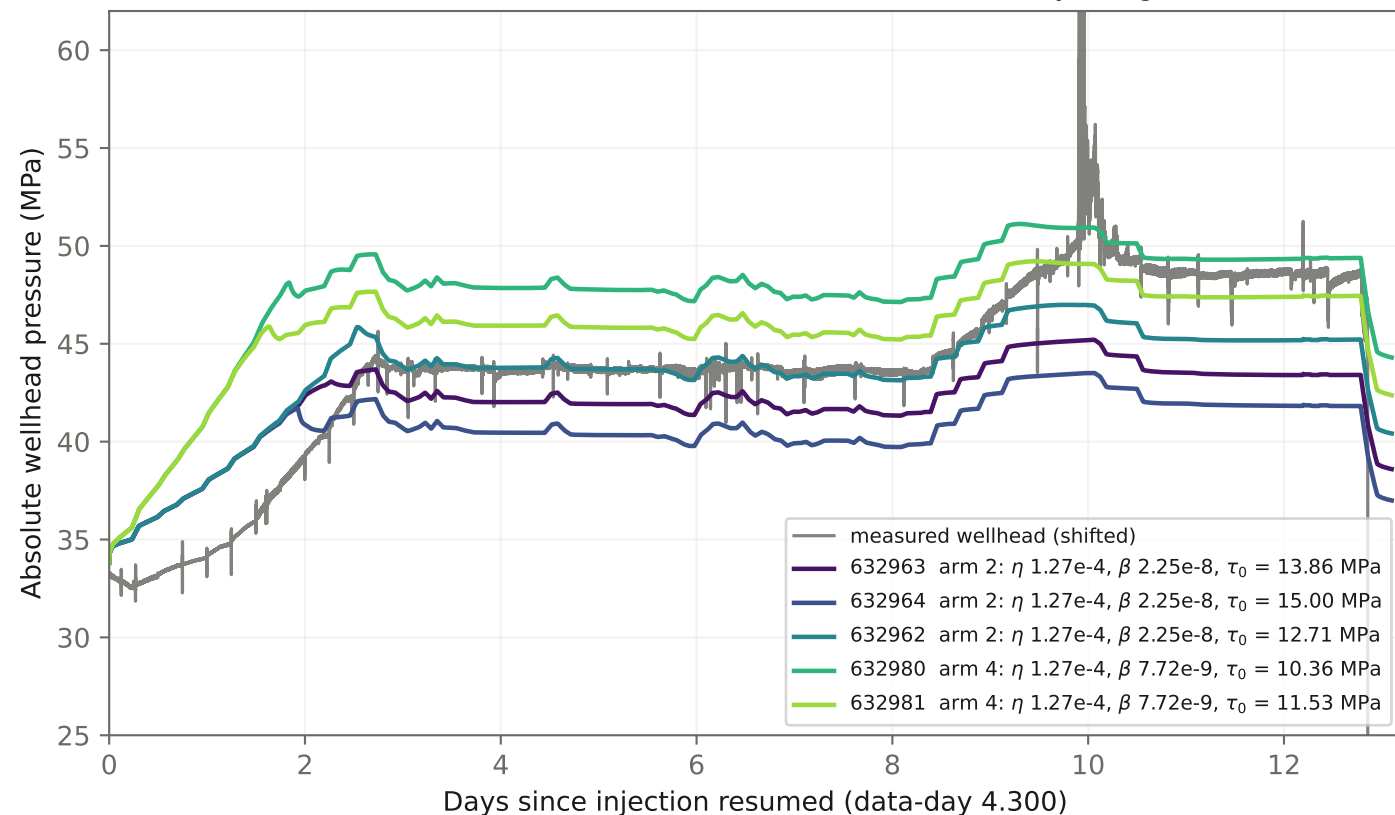
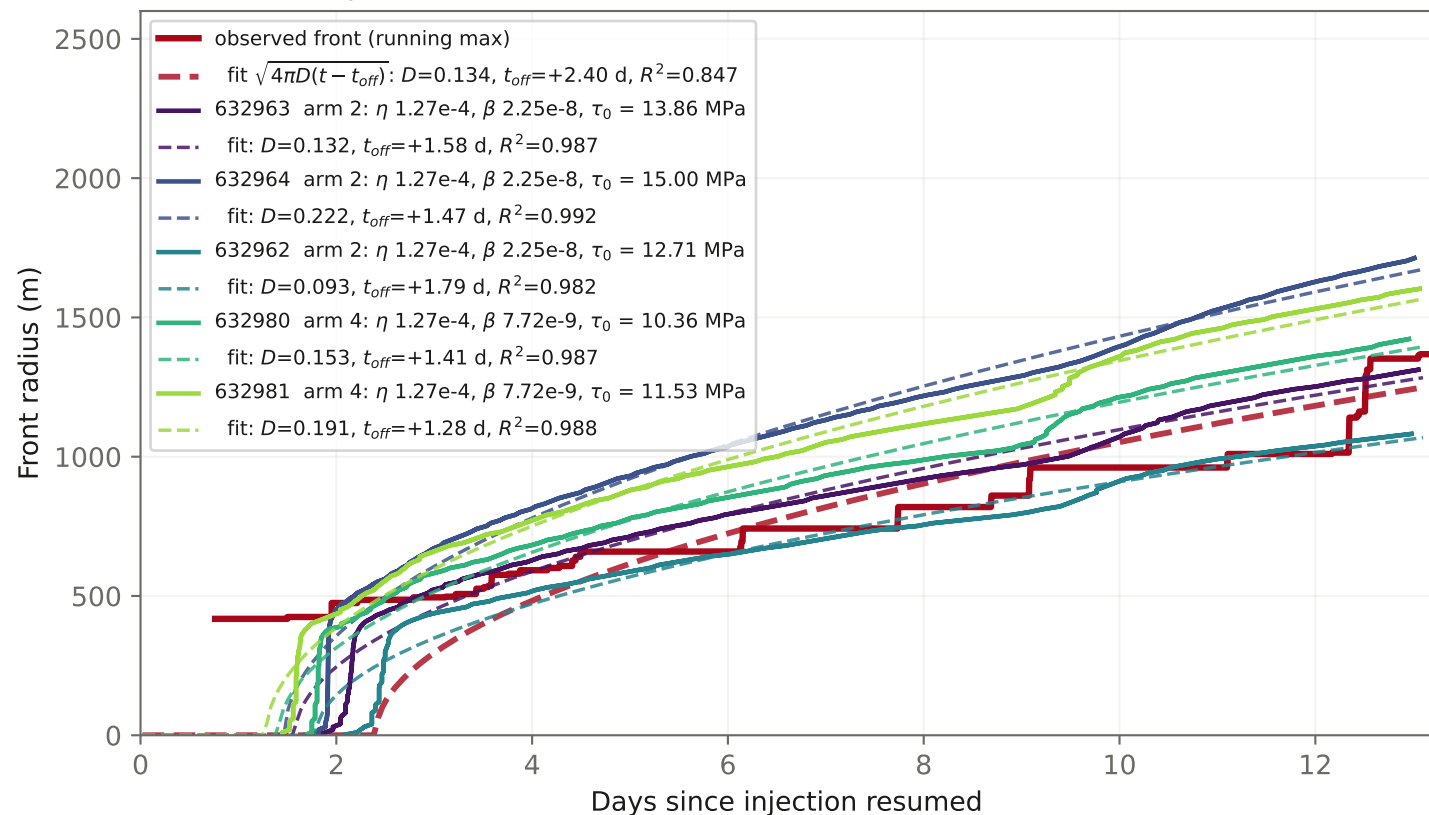


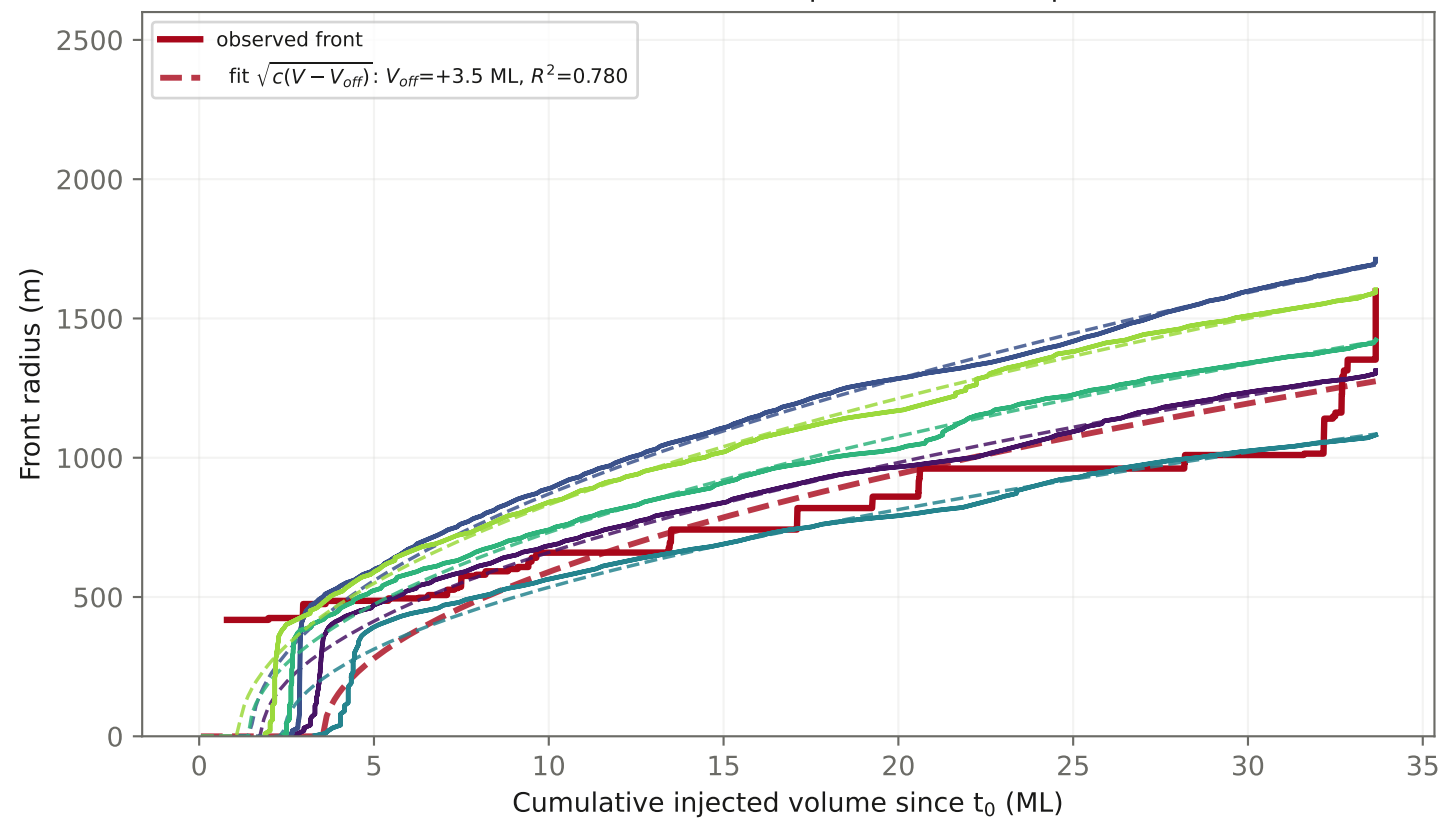
Wellhead pressure. Observed pulled back 4.300 d;
the 87.8 MPa water-hammer transient is off-scale by design



R-T with square-root fits (dashed).
 $r = \sqrt{4\pi D(t - t_{off})}$ — the $\sqrt{t}$ law with a shifted origin, since neither front starts at $r = 0, t = 0$



R-V with square-root fits (dashed).
Volume removes the rate-step staircase that spoils R-T



Slip vs the observed INCREMENT since t_0 .
HBI zeroes slip at restart, so absolute slip is not the target

